

Daily Integral - Medium

$$\int_0^{\pi/4} \left(1 + \frac{1 - 2\cos(x) + \cos(x)^2}{\sin(x)^2}\right)^{-1} \left(\frac{\tan(x)\cos(x)}{1 + \cos(x)}\right) dx$$

This problem was generously submitted by Ben S. on <https://dailyintegral.com>.

Solution

Firstly, factor $1 - 2\cos(x) + \cos(x)^2$:

$$\int_0^{\pi/4} \left(1 + \frac{(1 - \cos(x))^2}{\sin(x)^2}\right)^{-1} \left(\frac{\tan(x)\cos(x)}{1 + \cos(x)}\right) dx.$$

Now, use the Pythagorean identity for $\sin(x)^2$ and the quotient identity for $\tan(x)$ to get:

$$\int_0^{\pi/4} \left(1 + \frac{(1 - \cos(x))^2}{1 - \cos(x)^2}\right)^{-1} \left(\frac{\sin(x)}{1 + \cos(x)}\right) dx.$$

Now, factor $1 - \cos(x)^2$ using the difference of squares:

$$\begin{aligned} & \int_0^{\pi/4} \left(1 + \frac{(1 - \cos(x))^2}{(1 - \cos(x))(1 + \cos(x))}\right)^{-1} \left(\frac{\sin(x)}{1 + \cos(x)}\right) dx \\ &= \int_0^{\pi/4} \left(1 + \frac{1 - \cos(x)}{1 + \cos(x)}\right)^{-1} \left(\frac{\sin(x)}{1 + \cos(x)}\right) dx \\ &= \int_0^{\pi/4} \left(\frac{2}{1 + \cos(x)}\right)^{-1} \left(\frac{\sin(x)}{1 + \cos(x)}\right) dx \\ &= \int_0^{\pi/4} \frac{1 + \cos(x)}{2} \cdot \frac{\sin(x)}{1 + \cos(x)} dx \\ &= \frac{1}{2} \int_0^{\pi/4} \sin(x) dx \\ &= -\frac{1}{2} \cos(x) \Big|_0^{\pi/4} \\ &= -\frac{1}{2} \left(\frac{\sqrt{2}}{2} - 1\right) \\ &= \frac{2 - \sqrt{2}}{4}. \end{aligned}$$